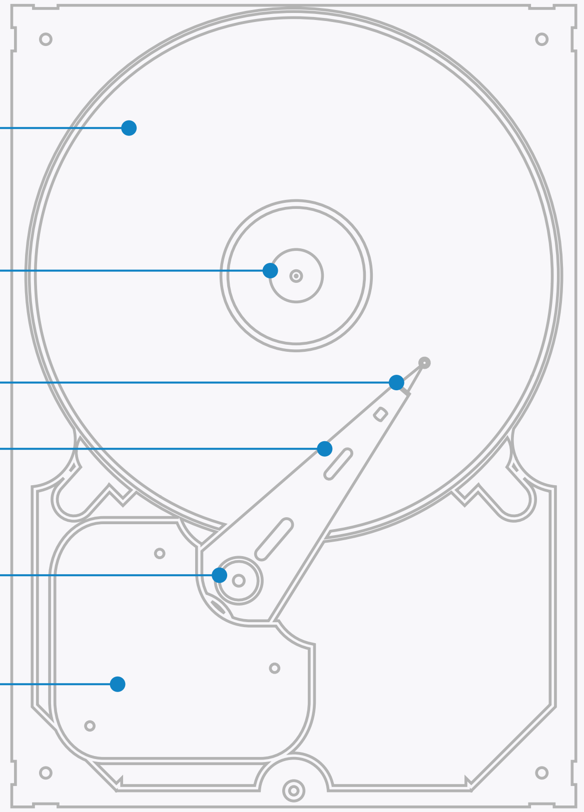
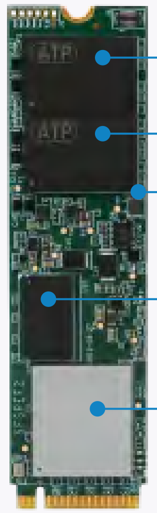
Introduction: What is NAND Flash Memory?

NAND flash memory is a type of non-volatile solid-state storage that persistently stores and retrieves data. It is non-volatile memory since it retains data when power is not applied.

Unlike hard disk drives, storage devices with flash memory have no mechanical moving parts — they don't have spinning platters or read/write heads. Instead, they store data in stationary NAND flash chips.

HDD vs SSD





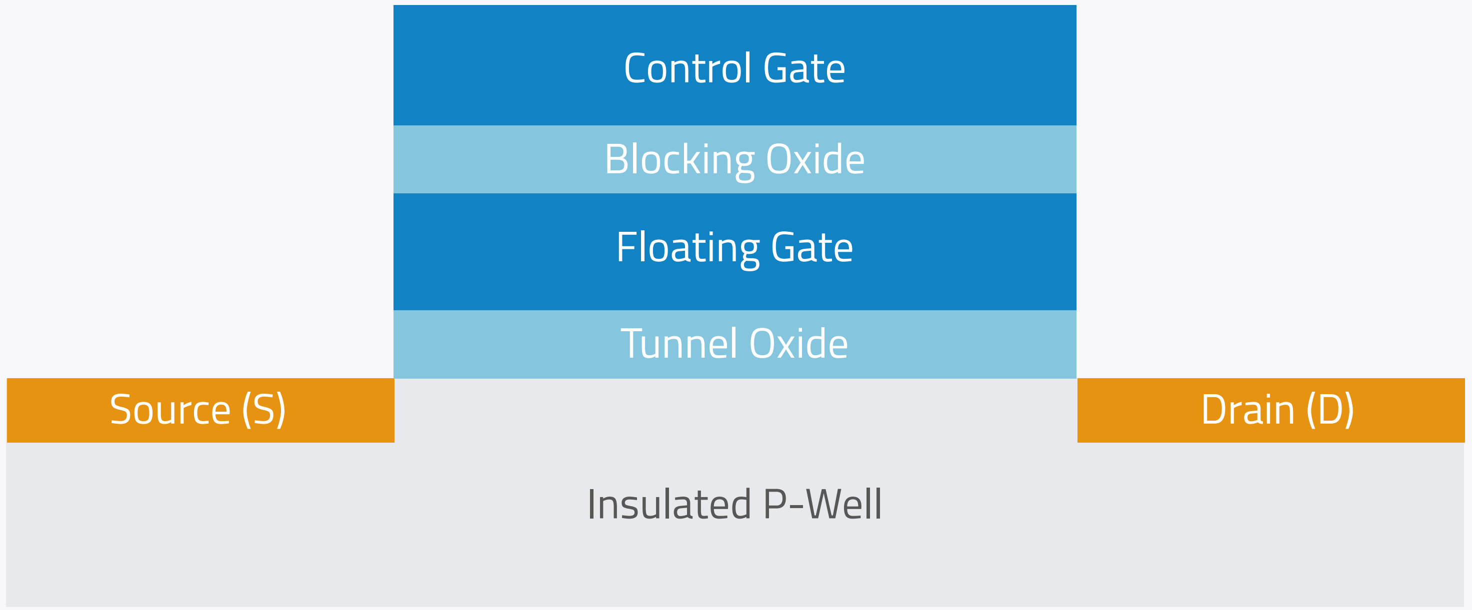
NAND Flash Cell Structure

The basic building block of flash memory is the NAND flash cell. The following diagram shows the main elements of a single NAND flash cell. Each cell contains a floating gate transistor, where the data are stored, through program/erase operations.

Oxide layers insulate the Floating Gate (storage layer), preventing the electrons to leak out of it.

Any electron stored in the FG is trapped, which means the cell is programmed (charged) and has a binary value of 0.

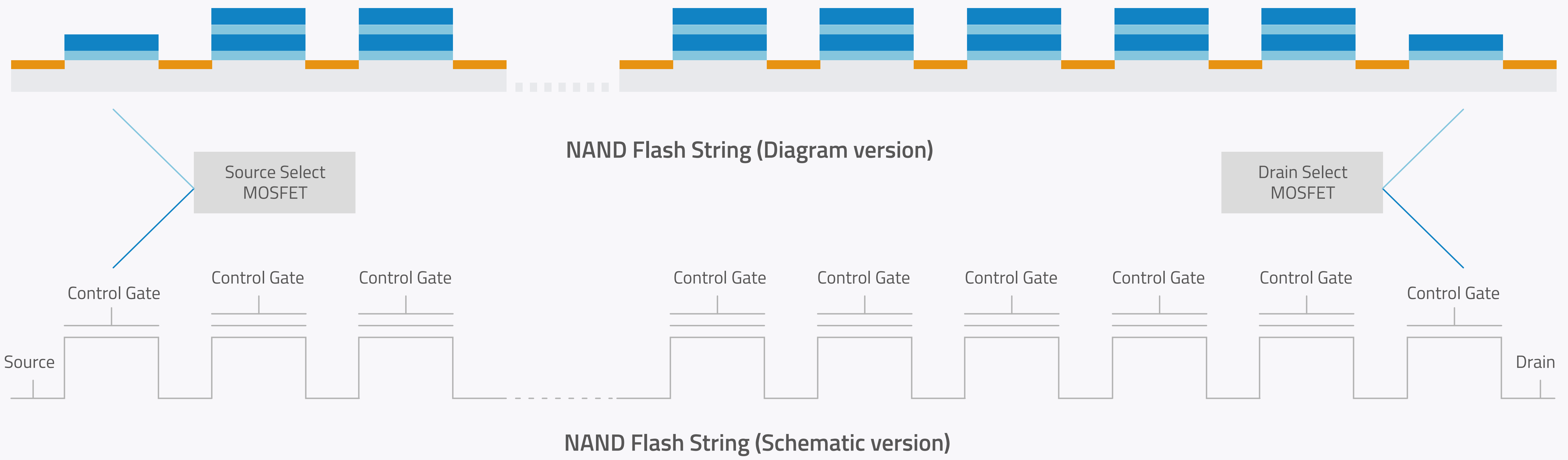
When the FG holds no charge, it has a binary value of 1.



NAND Flash Cell Structure

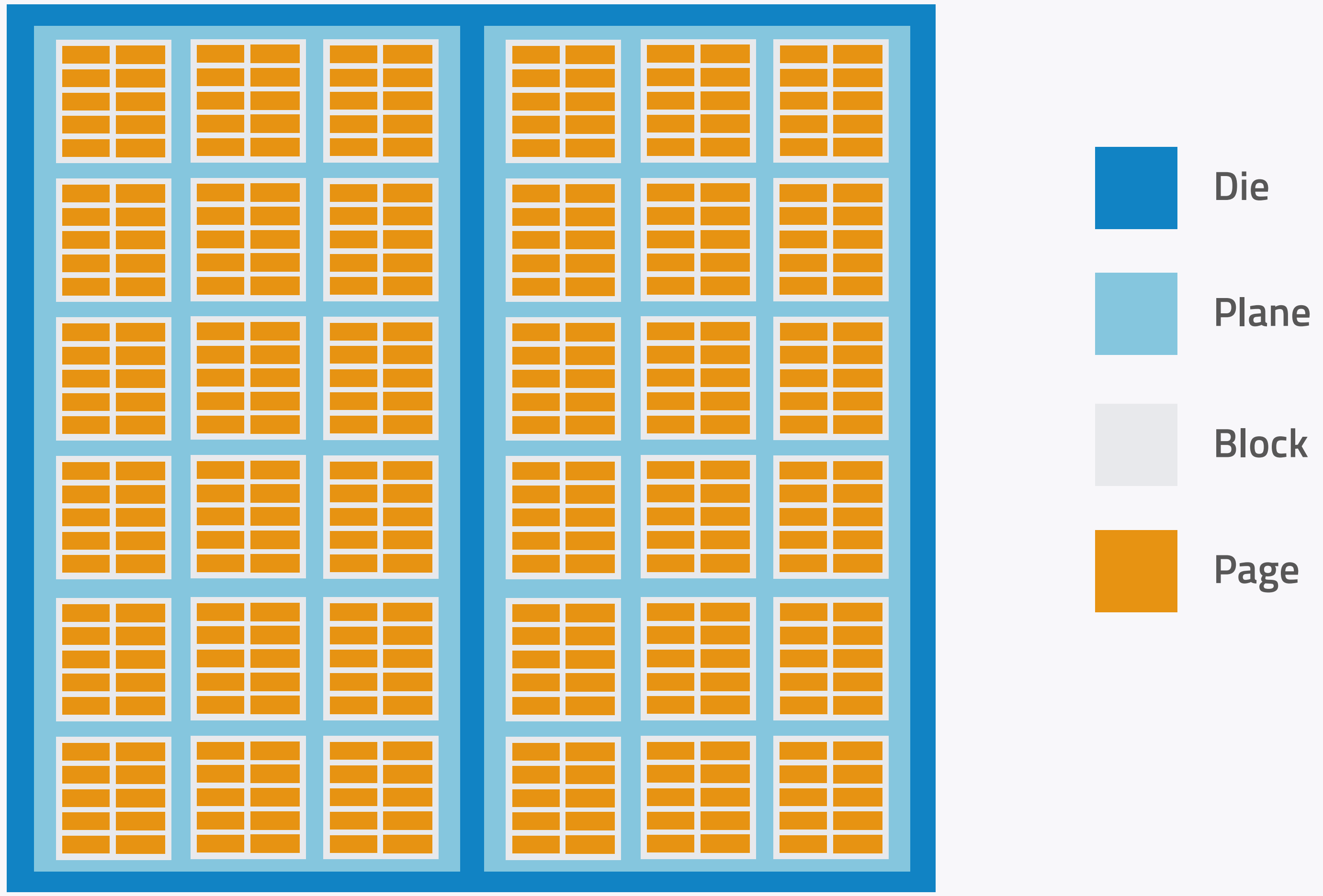
NAND Flash String

Several (32 to 128) NAND flash cells connected in series form a String, which is a quite compact structure. Several strings vertically arranged form a block.



NAND Flash Layout

A physical NAND **Page** is the group of NAND flash cells belonging to the same block, which share, horizontally, the same Control Gate (called Word Line). A NAND **Block** is composed of several pages. Several NAND blocks form a **Plane**. Finally, planes form a **Die**. Each memory chip contains one or more dies.

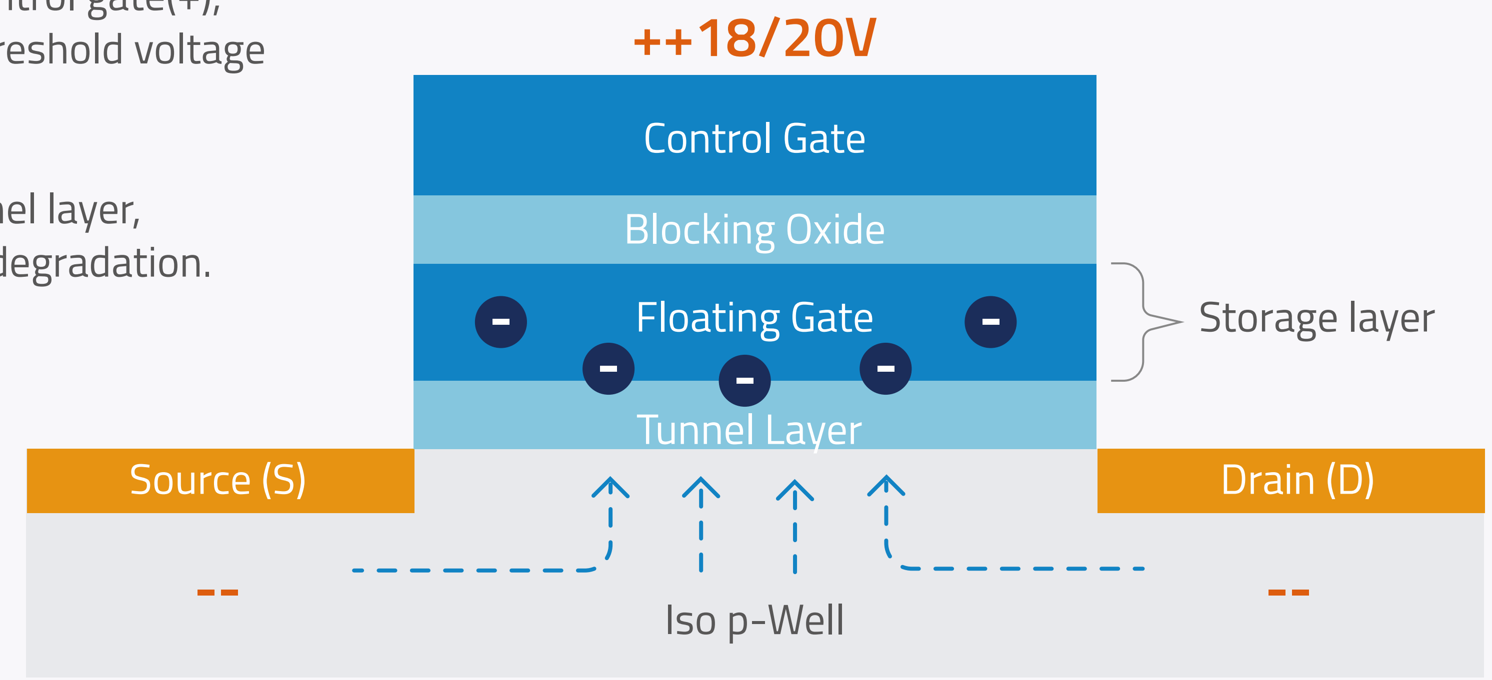


Basic Operations

Program (Write) Operation

A page is the smallest unit that can be programmed or written to, which is typically 8 to 16 KB in size. To write data, several positive high voltage (15V / 22V) pulses are applied to the control gate. After every pulse, a read-verify is performed. The program sequence will be completed once the verify will pass on all cells in the selected page. During each pulse, electrons move from the iso-p-well to the floating gate.

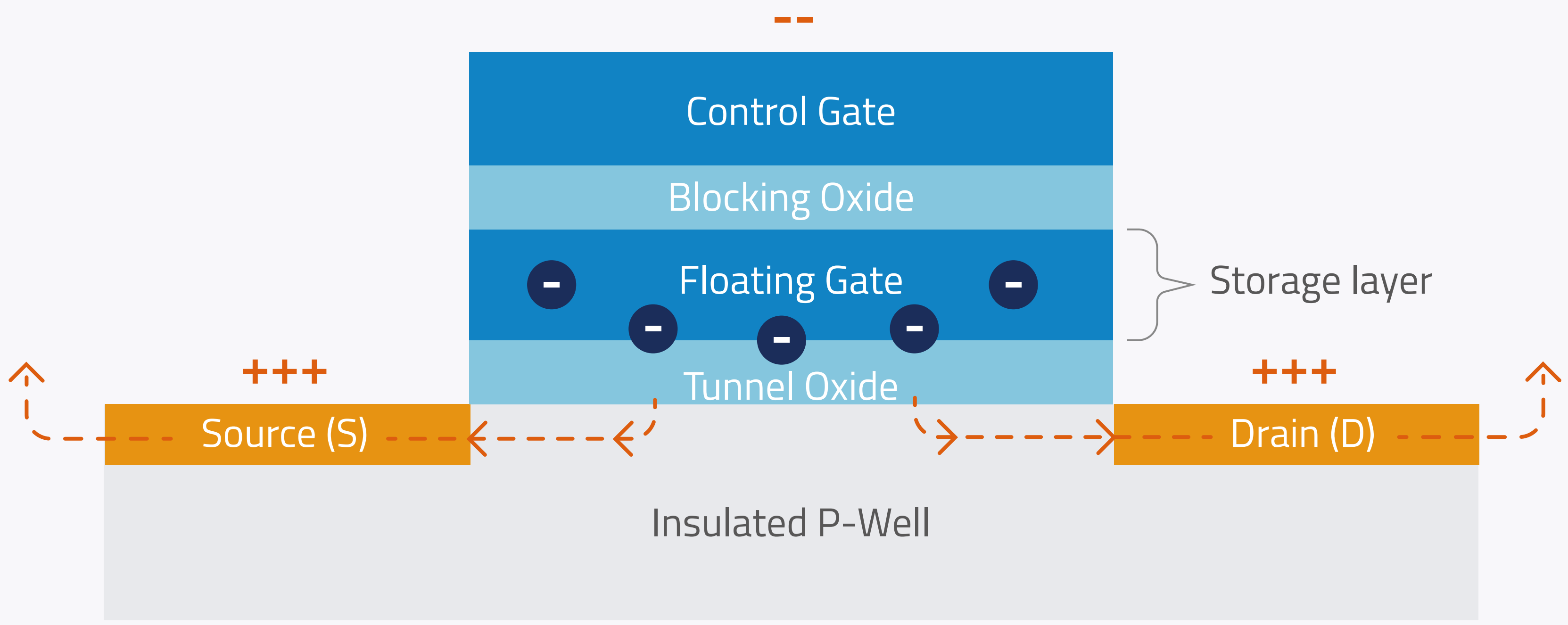
- Electrons are injected into the floating gate mainly through Fowler-Nordheim (FN) tunneling effect.
- FN tunneling requires a high electric field between the Source (-) and the control gate(+), which enable electrons to flow into the floating gate, thus increasing the threshold voltage of the NAND cell (V_t).
- The flow of high energy (Hot) carriers (electrons and holes) through the tunnel layer, during Program Operation, leads to oxide lattice damages and tunnel layer degradation.

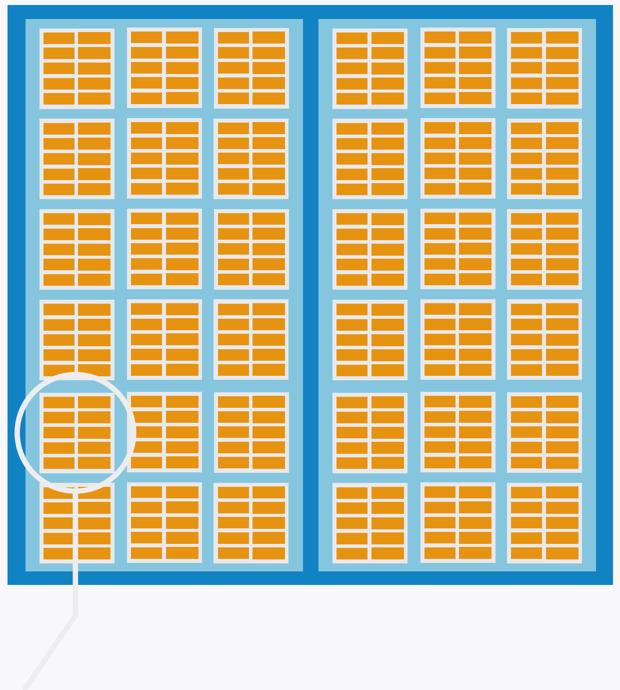


Basic Operations

Erase Operation

A block is the smallest NAND die portion that can be erased. To erase data, several negative high voltage (-15V / -22V) pulses are applied to the control gate. After every pulse, a read-verify is performed. The erase sequence will be completed once the verify will pass on all cells in the block.





- An erase operation clears the data from all pages in the block.
- If some pages contain active data, these are first copied to a spare block. The old block is then erased and ready to be written with new data.

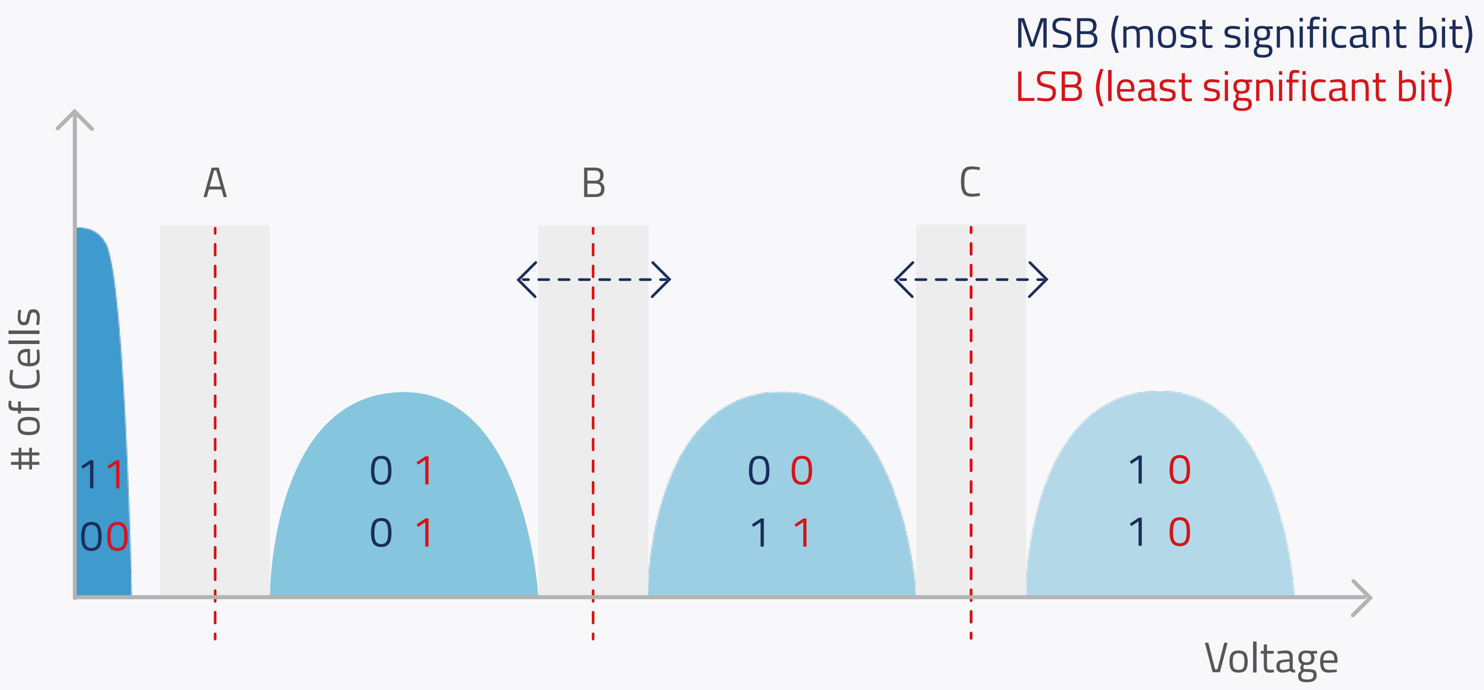
Basic Operations

Read Operation

A read operation is performed by comparing stored voltages with a threshold voltage (V_{th}). The maximum amount of data which can be read with a single read command from a NAND is a page. A page is the minimum read unit.

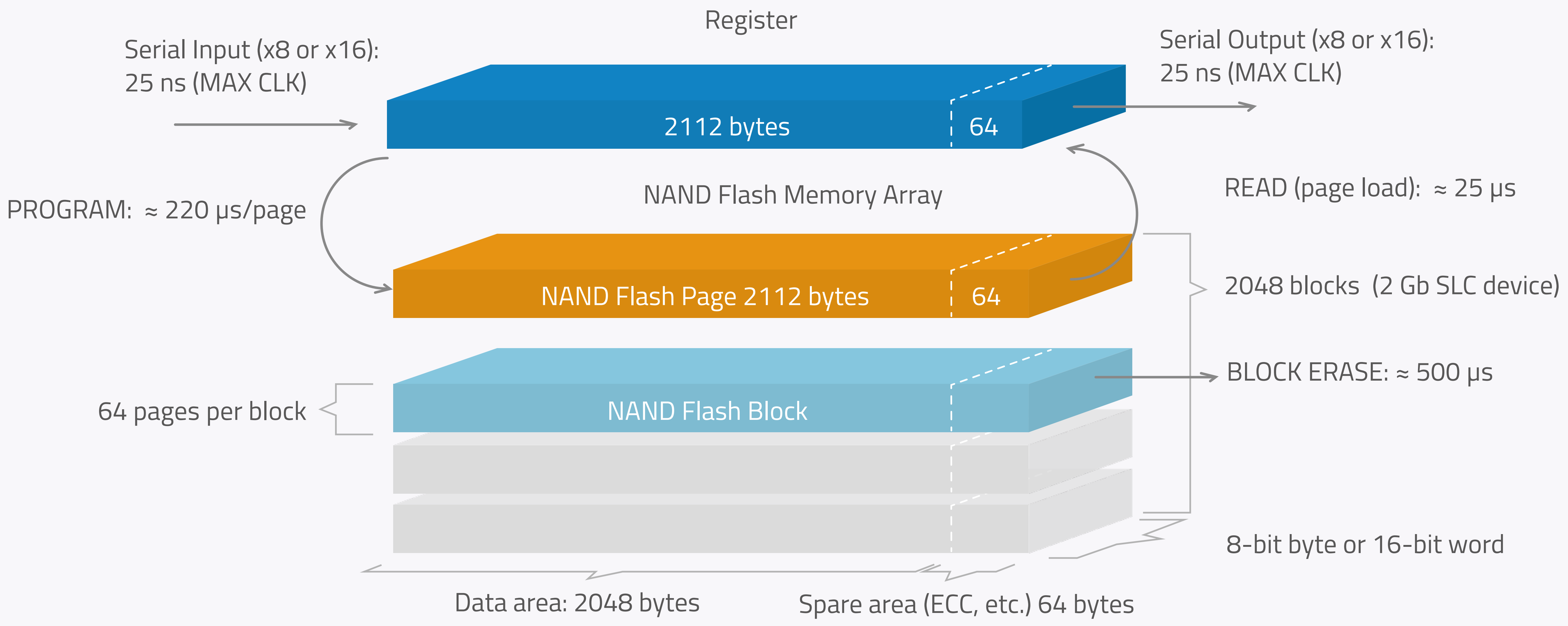
MLC (multi-level cell) flash can take four different voltage levels to store two bits and employs three different thresholds to separate the levels.

Example: MLC Read Thresholds



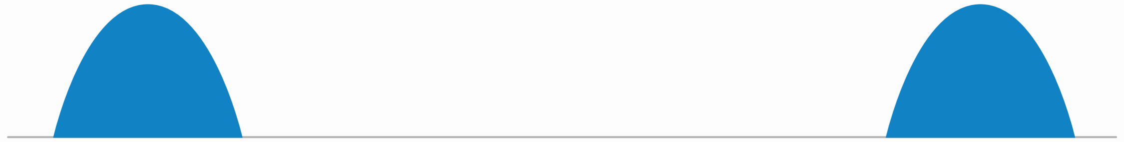
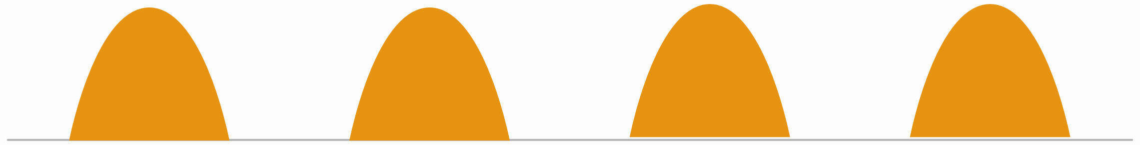
NAND Flash Layout

Here is an example of a 2 Gb flash device organized as 2048 blocks. The figure below shows how long erasing, programming and reading approximately takes.

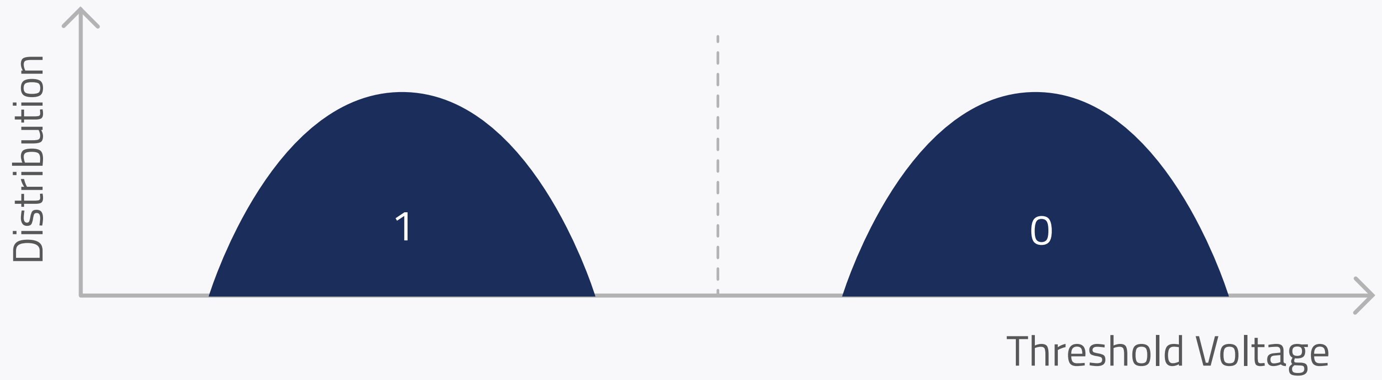


NAND Flash Types

Comparison Table

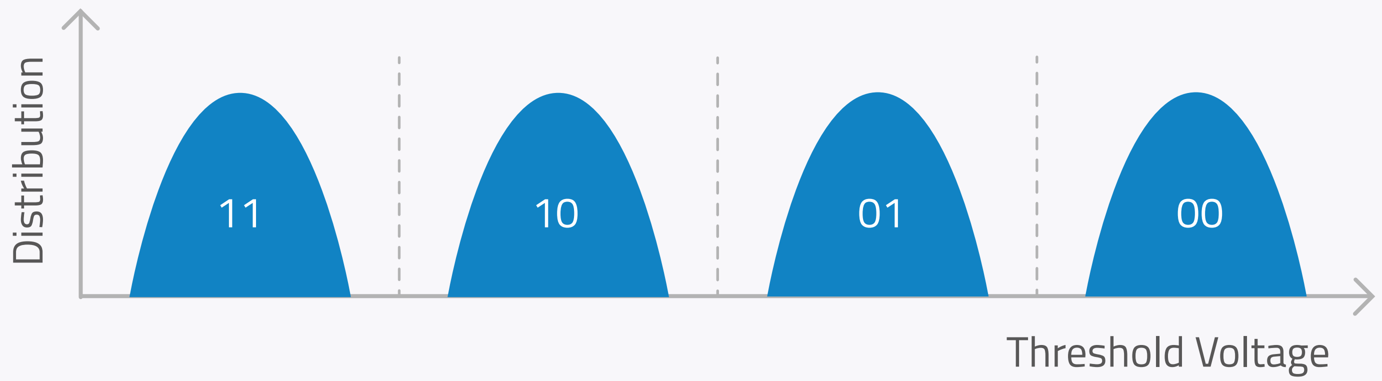
Technology Type	Definition	Endurance	Programming
SLC (Single-Level Cell)	Stores 1 bit per cell	50 to 100K P/E Cycles	
SLC Mode (Pseudo SLC / Advanced MLC)	▪ MLC flash that functions like SLC ▪ Stores 1 bit per cell instead of 2	20 to 50K P/E Cycles	
MLC (Multi-Level Cell)	Stores 2 bits per cell	3 to 5K P/E Cycles	
TLC (Triple-Level Cell)	Stores 3 bits per cell	~ 1K P/E Cycles	

NAND Flash Types and Threshold Voltage (V_{th}) Distribution



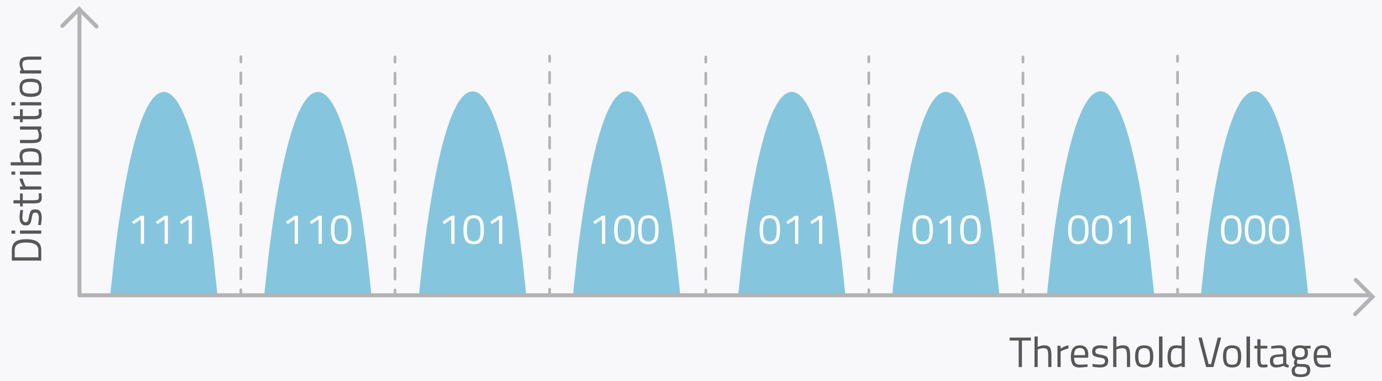
SLC

- High voltage margin makes cell reading easier and quicker.
- Small impact of leakage and cell interference
- Wider distribution of logic levels means programming or erasing at lower voltage for increased durability and longer product lifetime



MLC

- Lower cost per bit compared with SLC
- Closer distribution means voltage leakages can generate more impact and reduce product lifetime



TLC

- Lower cost per bit compared with MLC
- Very narrow voltage margin makes programming and reading speed very slow
- Much closer distribution magnifies wear and significantly reduces product lifetime